

Mid-Semester Exam: Discrete Mathematics (CSE121)

Date: 27-09-2025

Maximum marks: 60

Instructions. Please read the following points carefully.

- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- Write with either a black pen or a blue pen. Use of pencil is strictly prohibited.
- If you are found copying or talking to any other student, you will be disqualified from this exam, and the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or the invigilators.
- This exam is designed for 2 hours and 30 minutes.
- You can use the results given in the appendix without proving them. Apart from this, you can not assume any claim or result proven in the class/tutorial/quiz/homework. In such a case, you will have to prove things from scratch.

1 Problem Set

Question 1.1. 1. (3 marks). Let p, q, r be propositional variables. Write the truth tables of the following propositions:

(a) $p \implies (\neg q \oplus r) \iff (\neg p \vee q).$

(b) $p \implies \neg q \vee r \wedge \neg p \iff \neg r.$

2. (3 marks). Convert the following English statements first to logical expressions, and then write their negations first in logic and then in English. You should mention the rule of first-order logic you are using at every step. The algebra of propositional logic is given in the appendix.

(a) There exists a student of IIIT Delhi who knows singing or dancing.

(b) Every Indian citizen enjoys fundamental rights and bears fundamental duties.

3. (2 marks). Let x, y be variables. Are $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ equivalent for every predicate $P(x, y)$ and for every domains of x and y ? Justify your answer.

4. (2 marks). What do you understand by the inverse, converse, and contrapositive of an implication $p \implies q$? Which of these are equivalent to each other?

Question 1.2. 1. (1.5 marks). Let $p_1, \dots, p_n$ be propositional variables for some non-zero natural number n . Prove or disprove the following: There exists a propositional formula u in $p_1, \dots, p_n$ such that u is a tautology. Justify your answer by using the algebra of propositional logic given in the appendix.

2. (2.5 marks). Let p, q, r be propositional variables. Convert $(p \oplus q) \iff \neg r$ to an equivalent conjunctive form (need not be a conjunctive normal form) using only the algebra of propositional logic given in the appendix. Mention the rule used at every step.

3. (6 marks). Let $p_1, \dots, p_n$ be propositional variables. Let u_1 and u_2 be the propositional formulas defined as follows

$$u_1 := (p_1 \vee \neg p_2) \wedge (p_2 \vee \neg p_3) \wedge \dots \wedge (p_i \vee \neg p_{i+1}) \wedge (p_{i+1} \vee \neg p_{i+2}) \wedge \dots \wedge (p_{n-1} \vee \neg p_n) \wedge (p_n \vee \neg p_1).$$

$$u_2 := ((p_1 \wedge p_2 \wedge \dots \wedge p_{n-1} \wedge p_n) \oplus (\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n)) \wedge (p_1 \vee p_2 \vee \neg(p_1 \wedge p_2)).$$

Are u_1 and u_2 equivalent? Prove or disprove your answer.

Question 1.3. 1. (5 marks). Let a, b, c, d be integers such that $a + b + c = d$. Prove that d is odd if and only if either exactly one of a, b, c is odd or each of a, b, c is odd.

2. (3 marks). State whether the following is true or false. Justify your answers.

(a) Let x be a variable. Then, $x + 5 \geq 10$ is a proposition.

(b) Let $P(x)$ be the predicate ' $x + 5 \geq 10$ ', where the domain of x consists of all odd integers. Then, the negation of $\exists x P(x)$ is the following: for every even integer x , $x + 5 < 10$.

(c) For every prime number a, b satisfying $a < b$, there exists an odd natural number c such that $a < c < b$.

Question 1.4. Prove the following. Whenever you deal with integers, you are allowed to use only the properties of integers given in the appendix. You should mention which rule of integers you are using at every step in the proofs.

1. (3 marks). Let a be an integer. Prove that a^2 is odd if and only if $a + 3$ is even.

2. (3 marks). Let a, b be arbitrary integers. If $a^2 + b^2 = 0$ then prove that $a = 0$ and $b = 0$.

3. (4 marks). Let a, b, n be natural numbers. If $n = a \cdot b$ then prove that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Question 1.5. 1. (4 marks). Suppose there are finitely many prime numbers $p_1, \dots, p_n$ for some non-zero natural number n . Prove or disprove the following statement: there exists a natural number b , where $b \neq 1$ and $p_1 \cdots p_n + b$ is a prime number.

2. (6 marks). Prove that every collection of six people contains either a club of three people or a group of three strangers.

The Definition of club and stranger is given in the appendix (see Definition 1).

Question 1.6 (12 marks). For a natural number k , the notation $\sqrt[k]{2}$ means $2^{\frac{k}{k}}$.

Consider the collection of all numbers of the type $\sqrt[k]{2}$, where k is a non-zero natural number and k is a perfect square (Definition 2). Some elements of this collection are $\sqrt[1]{2}$, $\sqrt[4]{2}$, $\sqrt[16]{2}$, $\sqrt[36]{2}$, etc.

Are there finitely many irrational numbers in this collection? Prove or disprove your answer.

Appendix

A Definitions and results

Definition 1 (Club and strangers). *Let G be a group of people. If every pair of people in this group has met, then we call it a club. If every pair of people in G has not met, then we call it a group of strangers.*

Definition 2 (Perfect square). *An integer a is said to be perfect square if there exists an integer b such that $a = b^2$.*

Fact A.1. *Let a be a natural number which is not prime. Then, there exists a prime number b such that $b < a$ and $b \mid a$.*

B Important properties of integers

The following properties hold for every integers a, b, c .

1. **Closure under addition and multiplication:** $a + b, a \cdot b$ are integers.
2. **Associativity of addition:** $a + (b + c) = (a + b) + c$.
3. **Associativity of multiplication:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
4. **Commutativity of addition:** $a + b = b + a$.
5. **Commutativity of multiplication:** $a \cdot b = b \cdot a$.
6. **Distributive properties:** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.
7. **Identities:** $0 \neq 1, a + 0 = a, a \cdot 0 = 0$, and $a \cdot 1 = a$.
8. **Additive inverses:** There exists a unique integer m such that $m := -1 \cdot a$ and $a + m = 0$. The inverse of a is denoted $-a$.
9. **Subtraction:** $a - b$ is defined as $a + (-b)$.
10. **Cancellation:** If $a \cdot b = a \cdot c$ and $a \neq 0$ then $b = c$.
11. **No zero divisors:** If $a \cdot b = 0$ then either $a = 0$ or $b = 0$.
12. **Signed product:** $-a \cdot b = (-a) \cdot b = a \cdot (-b)$.

C Rules of the algebra of propositional logic

Let p, q, r be propositional variables, and T, F denote true and false. The rules are stated below.

1. **Rules for AND.**
 - (a) **Commutativity of AND:** $p \wedge q \equiv q \wedge p$.

- (b) **Associativity of AND:** $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
- (c) **Identity for AND:** $T \wedge p \equiv p$.
- (d) **Falsity for AND:** $F \wedge p \equiv p$.
- (e) **Idempotence for AND:** $p \wedge p \equiv p$.
- (f) **Contradiction for AND:** $p \wedge \neg p \equiv F$.

2. Rules for OR.

- (a) **Commutativity of OR:** $p \vee q \equiv q \vee p$.
- (b) **Associativity of OR:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
- (c) **Identity for OR:** $F \vee p \equiv p$.
- (d) **Truth for OR:** $T \vee p \equiv p$.
- (e) **Idempotence for OR:** $p \vee p \equiv p$.
- (f) **Validity for OR:** $p \vee \neg p \equiv T$.

3. Other rules.

- (a) **Double negation:** $\neg(\neg p) \equiv p$.
- (b) **Distributivity of AND over OR:** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- (c) **Distributivity of OR over AND:** $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- (d) **DeMorgan's laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.